

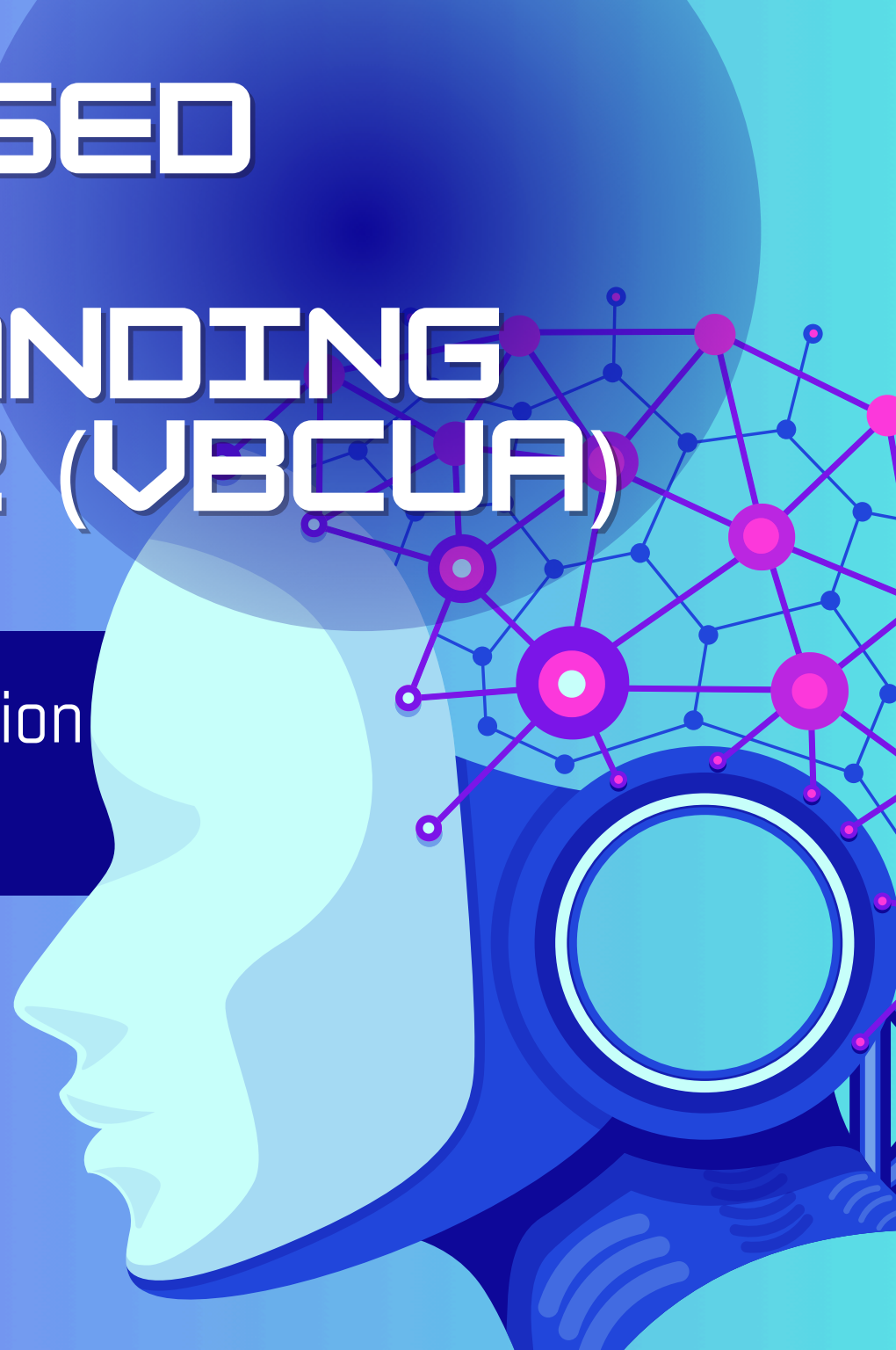
VOICE BASED CONCEPT UNDERSTANDING ANALYSER (VBCUA)

AI-Powered Speech Evaluation
System

Presented by:

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presenting to:
smart Bridge



Abstract

- Voice Based Concept Understanding Analyser (VBCUA) is an Artificial Intelligence based application developed to evaluate a student's conceptual understanding through spoken explanations. Traditional evaluation methods mainly focus on written examinations and manual assessment, which require significant time and effort.
- This project allows users to upload an audio explanation related to a selected topic. The application converts speech into text using OpenAI Whisper, compares the explanation with an ideal reference answer using Sentence-BERT semantic similarity, analyzes fluency through filler-word detection, extracts audio features such as duration and RMS energy using Librosa, and generates a final understanding score with feedback.
- The system is developed using Python and Streamlit, providing an easy-to-use web interface. It helps students improve communication skills and conceptual knowledge while assisting educators with faster evaluation.

OBJECTIVES:

- The primary objectives of the project are:
- To convert speech into text using AI.
- To evaluate conceptual understanding automatically.
- To compare user explanations with reference answers.
- To calculate semantic similarity using Sentence-BERT.
- To analyze speech fluency by detecting filler words.
- To extract audio characteristics using Librosa.
- To calculate a final understanding score.
- To generate an evaluation report.
- To provide instant feedback to users.

EXISTING SYSTEM & PROPOSED SYSTEM

Existing System

- Manual evaluation by teachers.
- Time-consuming assessment process.
- Subjective grading.
- No automatic speech analysis.
- No semantic understanding evaluation.

Proposed System

- The proposed Voice Based Concept Understanding Analyser automatically evaluates spoken explanations using Artificial Intelligence.
- Features include:
 - Speech-to-text conversion.
 - Semantic similarity analysis.
 - Fluency analysis.
 - Audio feature extraction.
 - Automatic scoring.
 - Instant feedback generation.
 - Report generation.

TECHNOLOGIES USED

Technology	Purpose
Python	Programming Language
Streamlit	Web Application
OpenAI Whisper	Speech Recognition
Sentence-BERT	Semantic Similarity
Librosa	Audio Feature Extraction
NumPy	Numerical Computation
Matplotlib	Waveform Visualization
ReportLab	Report Generation
Pydub	Audio Conversion

SYSTEM ARCHITECTURE

VOICE-BASED CONCEPT UNDERSTANDING ANALYSER (VBCUA)

SYSTEM FLOWCHART



MODULE DESCRIPTION

Modules

1. Speech-to-Text Module

- Converts uploaded speech into text using OpenAI Whisper.

2. Semantic Evaluation Module

- Calculates semantic similarity between the user's explanation and the reference answer using Sentence-BERT.

3. Fluency Analysis Module

- Counts filler words such as:
- um
- uh
- actually
- basically
- like

4. Audio Analysis Module

- Extracts:
- Duration
- RMS Energy
- Waveform
- using Librosa.

5. Scoring Engine

- Combines:
- Semantic Score
- Fluency Score
- to calculate the Final Understanding Score.

6. Report Generator

Generates a downloadable analysis report for the user.

Home Page

New tab

Voice Based Concept Understandir

+

localhost:8501

Chat

Deploy



Voice Based Concept Understanding Analyser

Upload your explanation audio and evaluate your conceptual understanding.

Select Topic

Deep Learning

Upload Audio File

 1783610...speech.mp3
343.2KB

+

0:00 / 0:14



Analyze Explanation

Activate Windows
Go to Settings to activate Windows.

30°C
Mostly cloudy

 Search







ENG
IN 09:02 PM
09-07-2026

Topic Selection

tab × Voice Based Concept Understanding × +

localhost:8501

 **Voice Based Concept Understanding Analyser**

Upload your explanation audio and evaluate your conceptual understanding.

Select Topic

Machine Learning

- Machine Learning
- Deep Learning
- Cloud Computing
- Python Programming
- Data Science
- Analyze Explanation

Activate Windows
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Audio Upload



Voice Based Concept Understanding Analyser

Upload your explanation audio and evaluate your conceptual understanding.

Select Topic

Deep Learning

Upload Audio File

1783610...speech.mp3 343.2KB

0:00 / 0:14

Analyze Explanation

Voice Based Concept Understanding



Voice Based Concept Understanding Analyser

Upload your explanation audio and evaluate your conceptual understanding.

Select Topic

Machine Learning

Upload Audio File

1783610...speech.mp3 361.2KB

0:00 / 0:15

Analyze Explanation

Transcription Completed!

Speech Transcription

Analyze Explanation

Transcription Completed!



Transcribed Text

Deep learning is a subset of machine learning that uses artificial neural networks with multiple hidden layers. These networks automatically learn complex patterns from large amounts of data. Deep learning is widely used in image recognition speech.



Evaluation Result

Semantic Score

89.31%

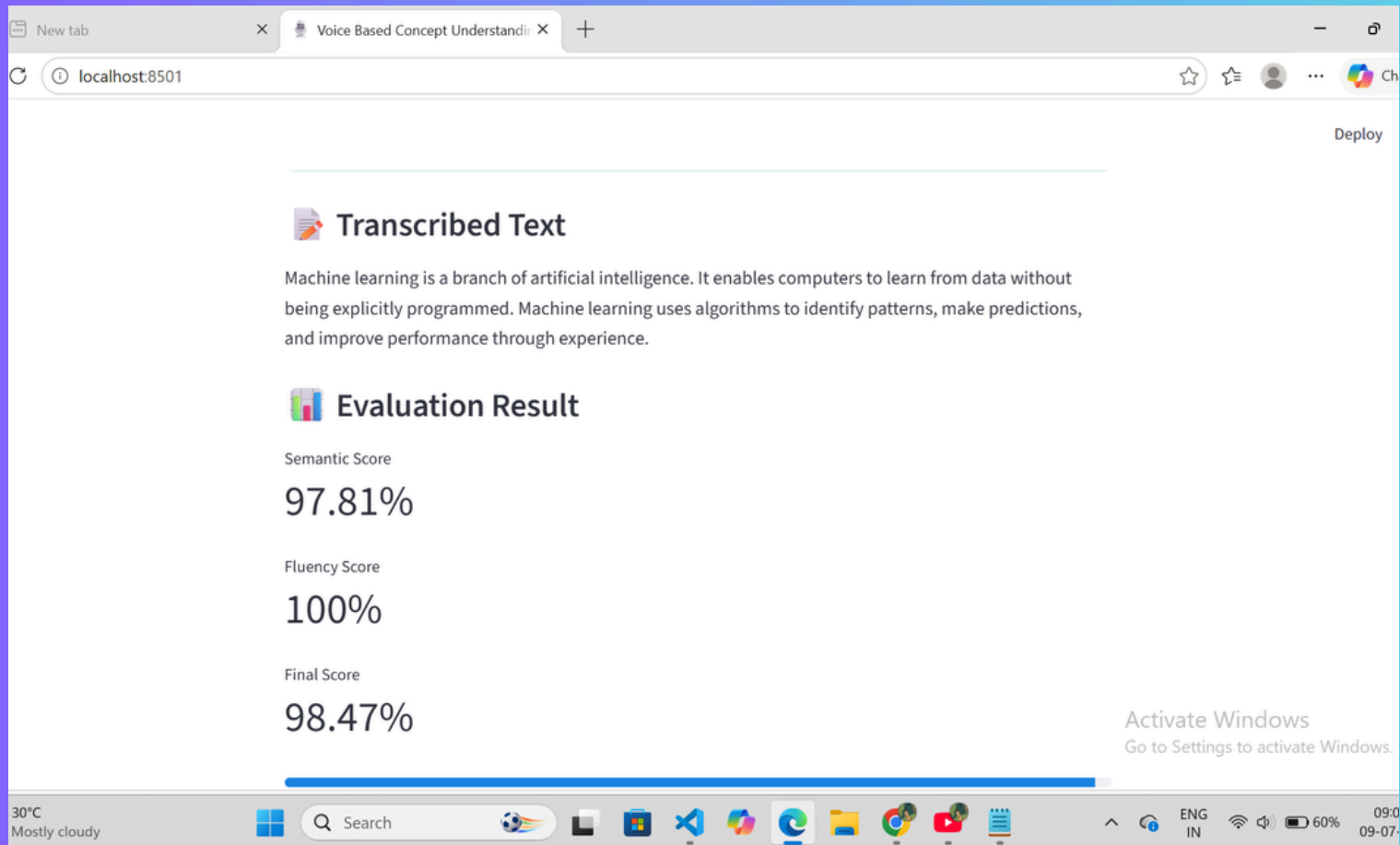
Fluency Score

100%

Final Score

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Evaluation Results



The screenshot shows a web browser window with two tabs: 'New tab' and 'Voice Based Concept Understanding'. The address bar shows 'localhost:8501'. The page content includes a 'Deploy' button in the top right corner. Below a horizontal line, there is a section titled 'Transcribed Text' with a document icon. The text below this title reads: 'Machine learning is a branch of artificial intelligence. It enables computers to learn from data without being explicitly programmed. Machine learning uses algorithms to identify patterns, make predictions, and improve performance through experience.' Below this is another section titled 'Evaluation Result' with a bar chart icon. This section displays three scores: 'Semantic Score' at 97.81%, 'Fluency Score' at 100%, and 'Final Score' at 98.47%. At the bottom right of the page, there is a Windows watermark that says 'Activate Windows Go to Settings to activate Windows.' The Windows taskbar is visible at the bottom, showing the search bar, task view button, and several application icons. The system tray on the right shows the date and time as '09:07' and '09-07'.

Deploy

Transcribed Text

Machine learning is a branch of artificial intelligence. It enables computers to learn from data without being explicitly programmed. Machine learning uses algorithms to identify patterns, make predictions, and improve performance through experience.

Evaluation Result

Semantic Score
97.81%

Fluency Score
100%

Final Score
98.47%

Activate Windows
Go to Settings to activate Windows.

30°C
Mostly cloudy

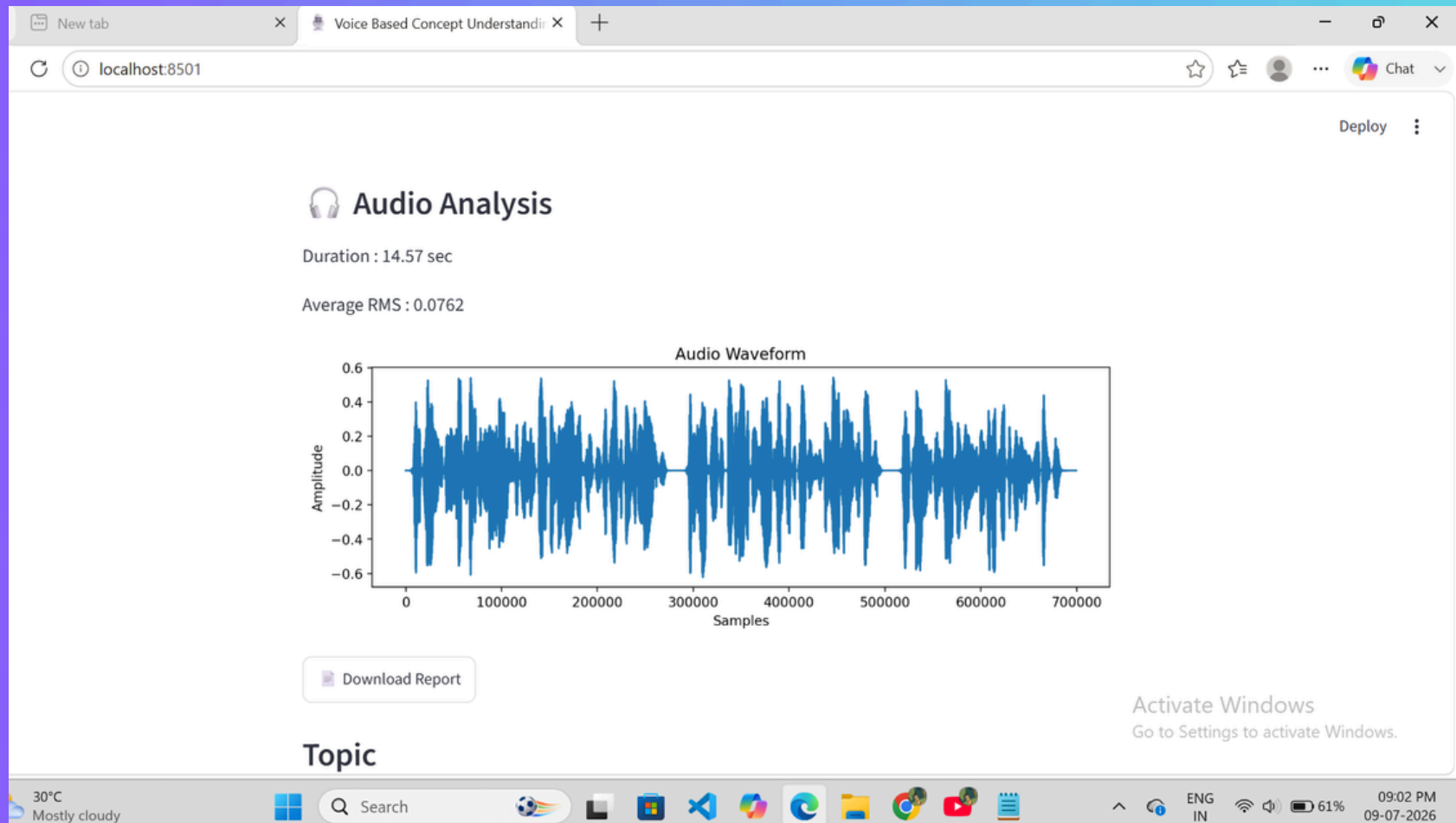
Search

ENG
IN

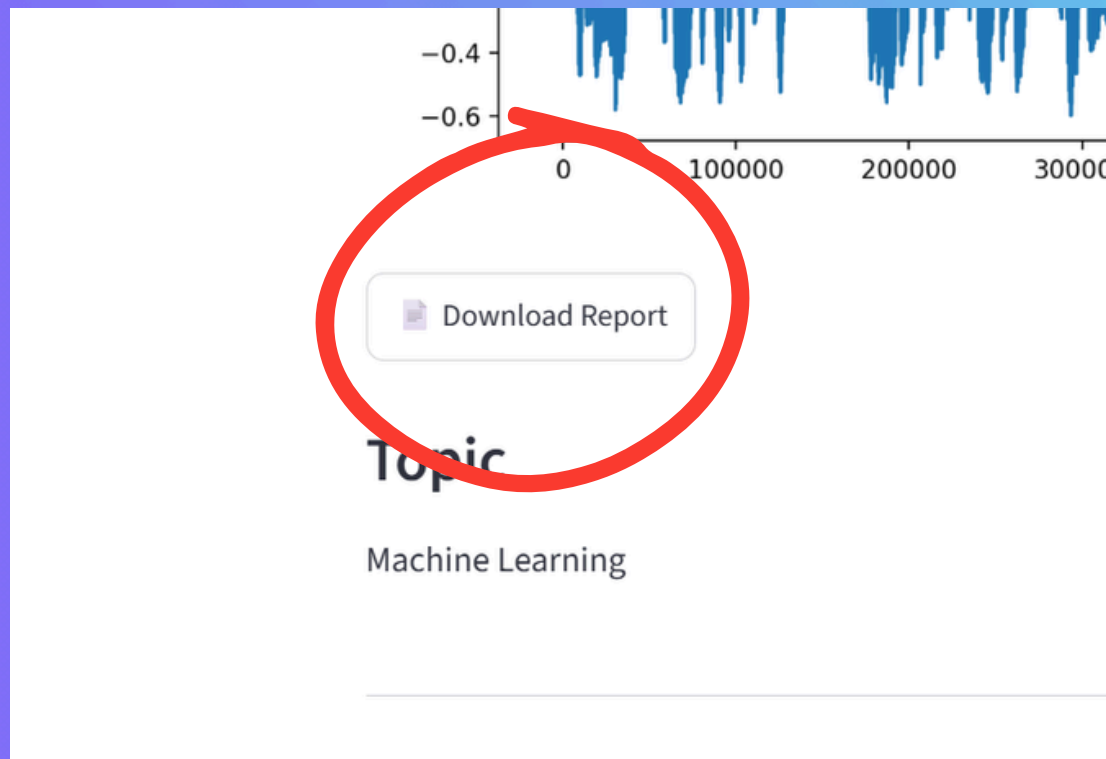
60%

09:07
09-07

Audio Analysis



Download Report Button



ADVANTAGES

- AI-powered evaluation
- Automatic speech recognition
- Fast assessment
- User-friendly interface
- High accuracy
- Instant feedback
- Interactive visualization
- Supports multiple technical topics

FUTURE SCOPE

- Multi-language support
- Real-time microphone input
- Emotion detection
- Pronunciation analysis
- Cloud deployment
- Teacher dashboard
- Student performance tracking
- Database integration

CONCLUSION

- The Voice Based Concept Understanding Analyser successfully demonstrates the application of Artificial Intelligence in educational assessment. By combining speech recognition, semantic similarity analysis, fluency evaluation, and audio feature extraction, the system provides an efficient method for evaluating conceptual understanding. The project reduces manual effort, provides immediate feedback, and enhances the learning experience for students. Future enhancements can further improve accuracy, scalability, and usability.

REFERENCES

1. OpenAI Whisper Documentation
2. Sentence-Transformers Documentation
3. Streamlit Documentation
4. Librosa Documentation
5. Python Official Documentation
6. ReportLab Documentation
7. NumPy Documentation
8. Matplotlib Documentation



THANK

YOU